Electronic Devices & Circuits Laboratory

Experiment 5: Closed Loop — Unity-Gain and Gain-of-10 Feedback Configurations

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October 3, 2026

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1 Aim

The objective of the experiment is to study the effect of negative feedback on the MOSFET differential amplifier. The same differential-amplifier core is surrounded by AC-coupled resistor feedback networks and operated in two configurations:

1. unity-gain configuration;
2. gain-of-10 configuration.

The measured closed-loop gain is compared with both the ideal resistor-ratio result and the finite-open-loop-gain prediction. The frequency response is also examined, with particular attention to the nominal RC corner frequency and the high-frequency roll-off.

2 Apparatus and Components

Table 1: Components and equipment used in the experiment.

S.No.	**Component / Equipment**	**Specification**	**Quantity**
1	Experiment 4 differential amplifier	SI9926BDY differential pair; $V_{DD} = 12\,\text{V}$, $R_D = 130\,\Omega$, $R_{SS} = 12\,\Omega$, $I_S \approx 80\,\text{mA}$	1
2	Feedback resistor R_f	$18\,\text{k}\Omega$, $\pm 5\%$	2
3	Input resistor R_{in}, unity gain	$18\,\text{k}\Omega$, $\pm 5\%$	2
4	Input resistor R_{in}, gain of 10	$1\,\text{k}\Omega$, $\pm 5\%$	2
5	Coupling capacitor C_f	$1\,\mu\text{F}$	2
6	Coupling capacitor C_{in}, unity gain	$1\,\mu\text{F}$	2
7	Coupling capacitor C_{in}, gain of 10	$10\,\mu\text{F}$	2
8	DC-bias isolation resistor R_{bias}	$1\,\text{M}\Omega$	2
9	Function generator	-	1
10	Oscilloscope	Two-channel measurement of differential input and output amplitudes and phase	1

3 Circuit and Feedback Topology

The prrevious Experiment MOSFET differential amplifier is used without changing its DC bias network. The feedback paths are AC-coupled so that the feedback network does

not alter the quiescent operating point. The two R_f branches are cross-coupled and the two input branches contain R_{in} and C_{in}.

A 1 MΩ resistor provides the DC path from each existing bias-potentiometer wiper to the corresponding gate, while the coupling capacitors isolate the AC feedback/input network from the DC bias. The crossed feedback connection provides negative feedback for the differential amplifier. An uncrossed connection would reverse the feedback sense.

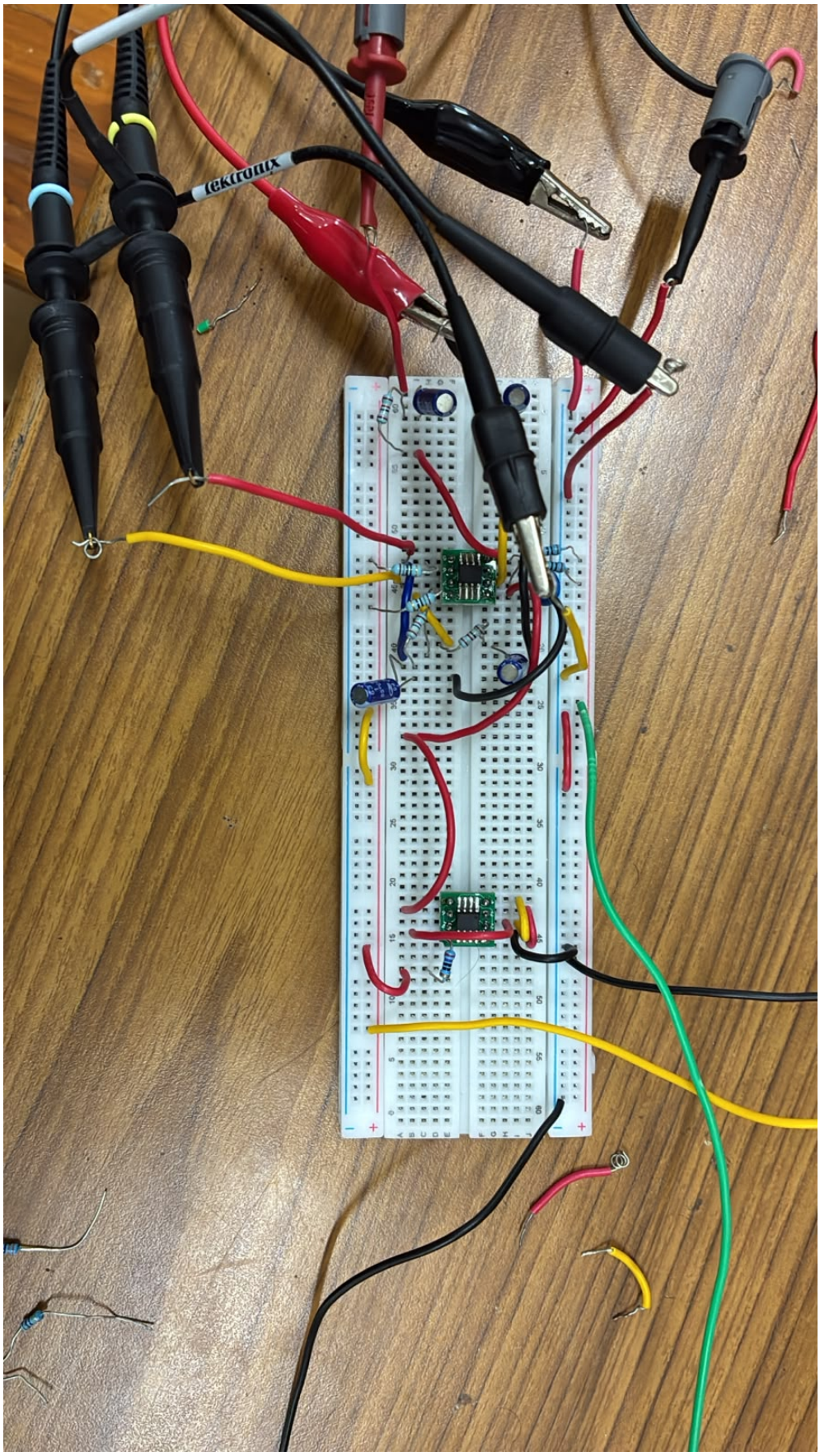


Figure 1: Circuit Diagram

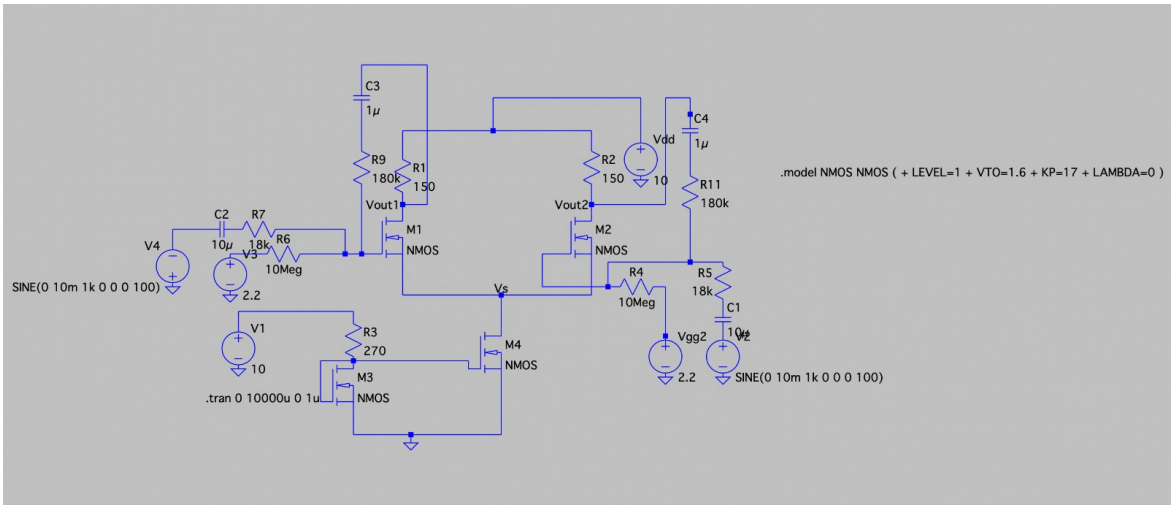


Figure 2: Cross-coupled negative feedback differential amplifier topology with AC coupling.

4 Theory

4.1 Differential Amplifier with Feedback

Let the differential source voltage be

$$V_{s,\text{diff}} = V_{s+} - V_{s-}, \tag{1}$$

and let the differential output be

$$V_{out,\text{diff}} = V_{out+} - V_{out-}. \tag{2}$$

For the differential pair used here, the crossed feedback connection is chosen so that the fed-back signal opposes the applied differential excitation. With the open-loop differential gain denoted by A_d, the laboratory manual gives the closed-loop relation

$$\frac{V_{out,\text{diff}}}{V_{s,\text{diff}}} = -\frac{A_d R_f}{R_f + R_{in}(1 + A_d)}. \tag{3}$$

Introducing the feedback factor

$$\beta = \frac{R_{in}}{R_f + R_{in}}, \tag{4}$$

Equation (3) can be written as

$$A_{v,\text{CL}} = -\frac{R_f/R_{in}}{1 + \dfrac{1}{\beta A_d}}. \tag{5}$$

The negative sign indicates inversion. Since the oscilloscope data used in this report are amplitude measurements, the measured tables use the gain magnitude

$$|A_v| = \frac{V_{out,pp}}{V_{in,pp}}. \tag{6}$$

The corresponding magnitude in decibels is

$$G_{\text{dB}} = 20 \log_{10} |A_v| . \tag{7}$$

4.2 Ideal Closed-Loop Gain

For sufficiently large loop gain ($\beta A_d \gg 1$), the correction term in Equation (5) becomes small and the ideal inverting-feedback result is obtained:

$$A_{v,\text{ideal}} = -\frac{R_f}{R_{in}}. \tag{8}$$

Thus, the resistor ratio determines the nominal closed-loop gain, while the finite value of A_d causes the actual magnitude to fall slightly below the ideal value in the simple feedback model.

4.3 Low-Frequency Corner Due to AC Coupling

Each series capacitor and its corresponding resistor form a first-order high-pass network. The nominal corner frequency is

$$f_c = \frac{1}{2\pi RC}. \tag{9}$$

For the unity-gain configuration,

$$RC = (18\,\text{k}\Omega)(1\,\mu\text{F}) = 18 \times 10^3 \times 10^{-6} = 1.8 \times 10^{-2}\,\text{s}. \tag{10}$$

Therefore,

$$f_{c,\text{unity}} = \frac{1}{2\pi(1.8 \times 10^{-2})} = 8.8419\,\text{Hz} \approx 8.84\,\text{Hz}. \tag{11}$$

For the gain-of-10 configuration, taking the resistance as $R = 18\,\text{k}\Omega$ and the coupling capacitance as $C = 10\,\mu\text{F}$,

$$RC = (18\,\text{k}\Omega)(10\,\mu\text{F}) = 18 \times 10^3 \times 10^{-5} = 1.8 \times 10^{-1}\,\text{s}. \tag{12}$$

Hence,

$$f_{c,\text{gain10}} = \frac{1}{2\pi(1.8 \times 10^{-1})} = 0.8842\,\text{Hz} \approx 0.884\,\text{Hz}. \tag{13}$$

Thus, the nominal low-frequency corners are

$$\boxed{f_{c,\text{unity}} \approx 8.84\,\text{Hz}} \tag{14}$$

and

$$\boxed{f_{c,\text{gain10}} \approx 0.884\,\text{Hz}}. \tag{15}$$

5 Experimental Measurement

The bias network was retained while the feedback network was added. In the unity-gain configuration, $R_f = 10\,\text{k}\Omega$, $R_{in} = 10\,\text{k}\Omega$, $C_f = 1\,\mu\text{F}$, and $C_{in} = 1\,\mu\text{F}$. The resistor associated with the input AC-coupling network was $18\,\text{k}\Omega$. In the gain-of-10 configuration, R_f and C_f were unchanged, while R_{in} was changed to $1\,\text{k}\Omega$ and C_{in} to $10\,\mu\text{F}$. The same $18\,\text{k}\Omega$ coupling-network resistor was retained.

The gain calculations below use peak-to-peak amplitudes, so the voltage units cancel in the ratio. The measured unity-gain input was $100\,\mathrm{mV_{pp}}$, while the measured gain-of-10 input was $40\,\mathrm{mV_{pp}}$.

The bias point was first checked with the feedback network disconnected. The unity-gain feedback network was then connected with $R_f = 10\,\mathrm{k\Omega}$, $R_{in} = 10\,\mathrm{k\Omega}$, and $1\,\mu\mathrm{F}$ coupling capacitors. The $18\,\mathrm{k\Omega}$ resistor in the AC-coupling network was retained. After the input and output amplitudes were recorded, the frequency was swept over the measured range. The gain-of-10 configuration was obtained by replacing R_{in} with $1\,\mathrm{k\Omega}$ and C_{in} with $10\,\mu\mathrm{F}$ while keeping R_f, C_f, and the $18\,\mathrm{k\Omega}$ coupling-network resistor unchanged. The same amplitude and frequency measurements were then performed for the gain-of-10 configuration.

6 Observations

6.1 Closed-Loop Gain at 1 kHz and at the Measured Peak Region

Table 2: Measured closed-loop gain at 1 kHz and at the maximum measured response region.

Configuration	$V_{in,pp}$ **(mV)**	$V_{out,pp}$ **(mV)**	$\lvert A_v \rvert$	**Gain (dB)**
Unity gain, 1 kHz	200	224	1.1200	0.97
Unity gain, peak measured region	228	228	1.0000	0.00
Gain of 10, 10 Hz	40	396	9.9000	19.91

6.2 Unity-Gain Frequency Response

The unity-gain frequency sweep was recorded with $V_{in,pp} = 200\,\mathrm{mV}$.

Table 3: Unity-gain low-frequency sweep.

Frequency (Hz)	$V_{out,pp}$ **(mV)**	$\lvert A_v \rvert$	**Gain (dB)**
1.0	172	0.8600	-1.31
1.1	180	0.9000	-0.92
1.2	188	0.9400	-0.54
1.3	188	0.9400	-0.54
1.4	196	0.9800	-0.18
1.5	196	0.9800	-0.18
1.6	204	1.0200	0.17
1.7	204	1.0200	0.17
1.8	204	1.0200	0.17
1.9	208	1.0400	0.34

Frequency (Hz)	$V_{out,pp}$ **(mV)**	$\lvert A_v \rvert$	**Gain (dB)**
2.0	212	1.0600	0.51
2.5	216	1.0800	0.67
3.0	220	1.1000	0.83
3.5	220	1.1000	0.83
4.0	220	1.1000	0.83
4.5	220	1.1000	0.83
5.0	224	1.1200	0.98
5.5	228	1.1400	1.14
6.0	228	1.1400	1.14
6.5	228	1.1400	1.14
7.0	228	1.1400	1.14
7.5	228	1.1400	1.14
8.0	228	1.1400	1.14
8.5	228	1.1400	1.14
9.0	228	1.1400	1.14
9.5	228	1.1400	1.14
10.0	224	1.1200	0.98

Table 4: Unity-gain main frequency sweep.

Freq. (Hz)	V_{out} **(mV)**	$\lvert A_v \rvert$	**dB**	**Freq. (Hz)**	V_{out} **(mV)**	$\lvert A_v \rvert$	**dB**
10	224	1.1200	0.98	22000	192	0.9600	-0.35
20	224	1.1200	0.98	24000	184	0.9200	-0.72
50	224	1.1200	0.98	26000	180	0.9000	-0.92
500	224	1.1200	0.98	28000	176	0.8800	-1.11
1000	224	1.1200	0.98	30000	168	0.8400	-1.51
2000	224	1.1200	0.98	32000	164	0.8200	-1.73
5000	224	1.1200	0.98	34000	160	0.8000	-1.94
8000	224	1.1200	0.98	36000	156	0.7800	-2.16
8200	220	1.1000	0.83	38000	152	0.7600	-2.38
8500	220	1.1000	0.83	40000	144	0.7200	-2.85
9000	216	1.0800	0.67	42000	140	0.7000	-3.10
10000	216	1.0800	0.67	45000	136	0.6800	-3.35
12000	212	1.0600	0.51	50000	124	0.6200	-4.15
15000	208	1.0400	0.34	56000	116	0.5800	-4.74
18000	200	1.0000	0.00	64000	104	0.5200	-5.68
20000	196	0.9800	-0.18	76000	88	0.4400	-7.13

6.3 Unity gain - Output from Oscilloscope

Note : The yellow and blue curves in the oscilloscope are the V_{out1} and V_{out2} respectively (the drain voltages of the mosfets). The red/pink colour sin waves are the final $V_{out1} - V_{out2}$ to be measured(differential outputs).

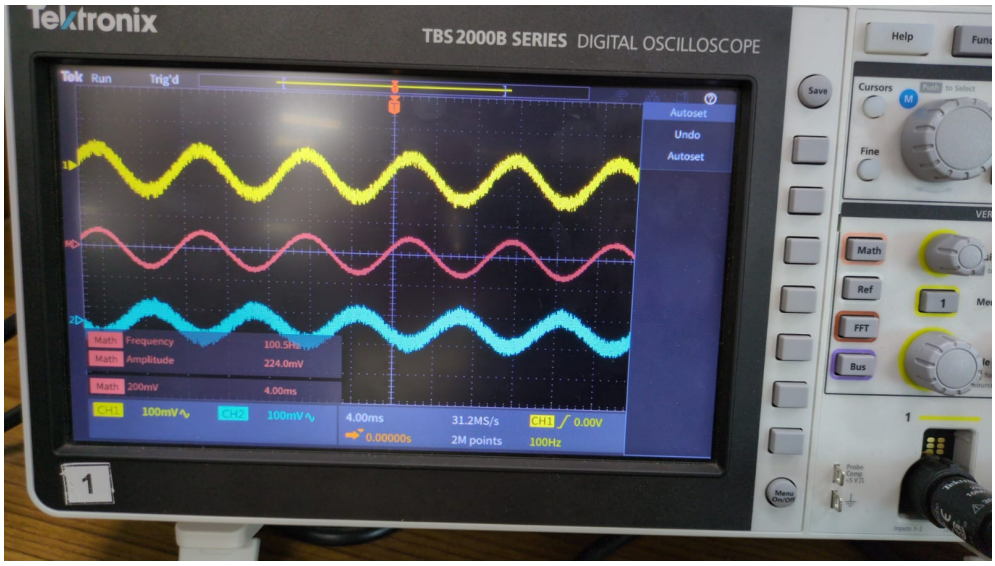


Figure 3: 100 Hz

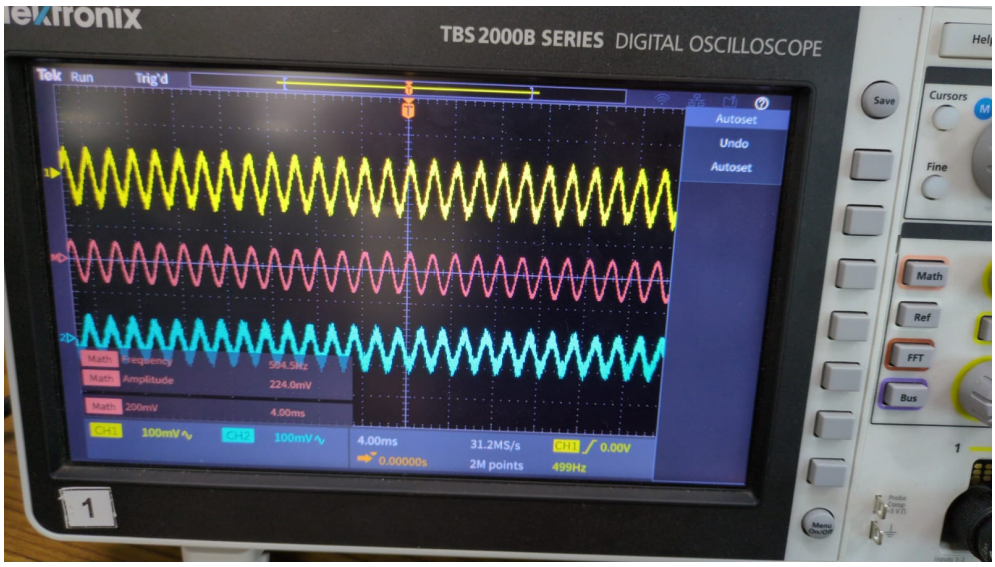


Figure 4: 500 Hz



Figure 5: 1 kHz

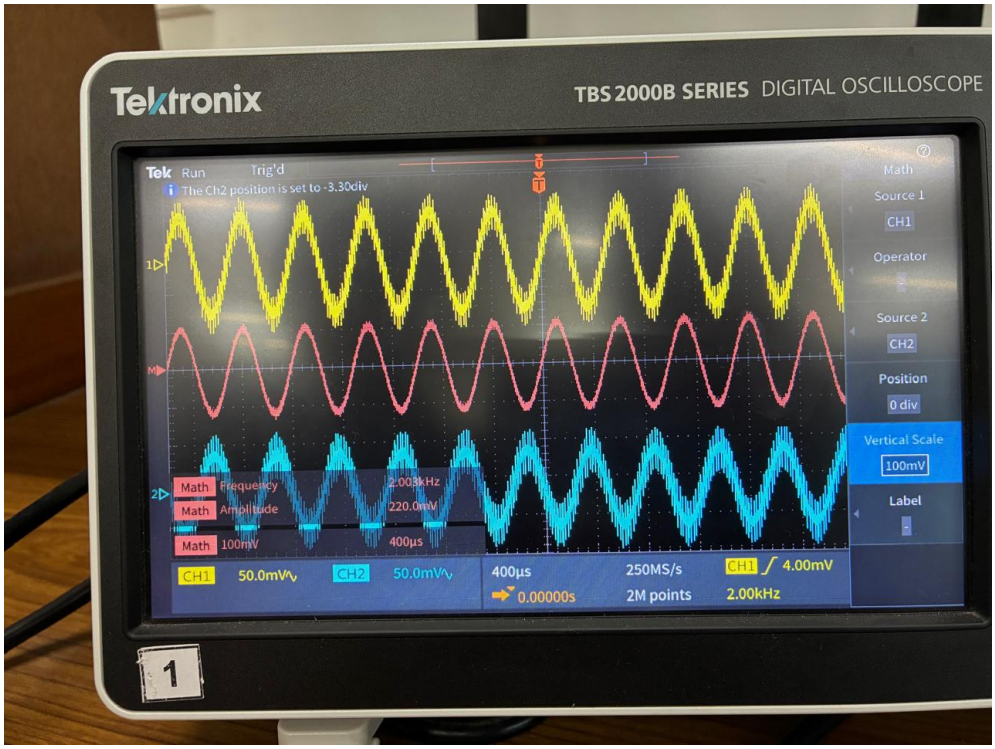


Figure 6: 2 kHz



Figure 7: 7 kHz

6.4 Gain-of-10 Frequency Response

The gain-of-10 frequency sweep was recorded with $V_{in,pp} = 40\,\text{mV}$.

Table 5: Gain-of-10 frequency sweep.

Frequency (Hz)	$V_{out,pp}$ **(mV)**	$\lvert A_v \rvert$	**Gain (dB)**
1	312	7.8000	17.84
5	352	8.8000	18.89
7	396	9.9000	19.91
9	396	9.9000	19.91
10	396	9.9000	19.91
15	388	9.7000	19.74
50	372	9.3000	19.37
100	366	9.1500	19.23
200	366	9.1500	19.23
500	332	8.3000	18.38
700	332	8.3000	18.38
1000	312	7.8000	17.84
1100	307	7.6750	17.70
1200	300	7.5000	17.50
1500	288	7.2000	17.15
1700	272	6.8000	16.65
2000	255	6.3750	16.09
5000	180	4.5000	13.06

7 Graphs and Frequency Response Curves

- **Bode Magnitude Plot:** Closed-loop gain in dB vs. frequency ($\log X$ axis) plotted for both Unity-Gain and Gain-of-10 configurations.
- **Bandwidth Comparison:** Identification of low-frequency -3 dB corners ($f_{c,\text{low}}$) and high-frequency gain roll-off ($f_{c,\text{high}}$).

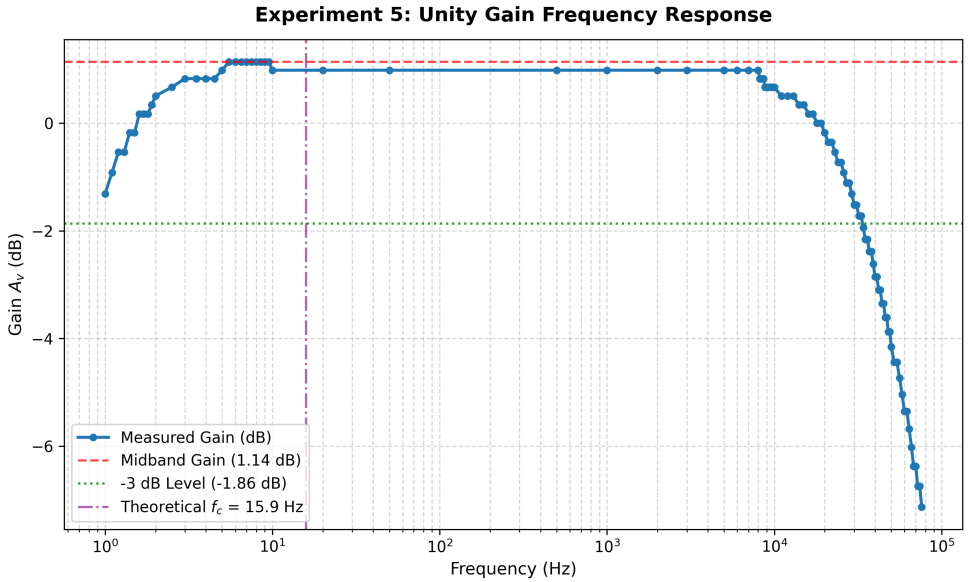


Figure 8: Frequency response curves (Bode magnitude plot) for Unity-Gain setup.

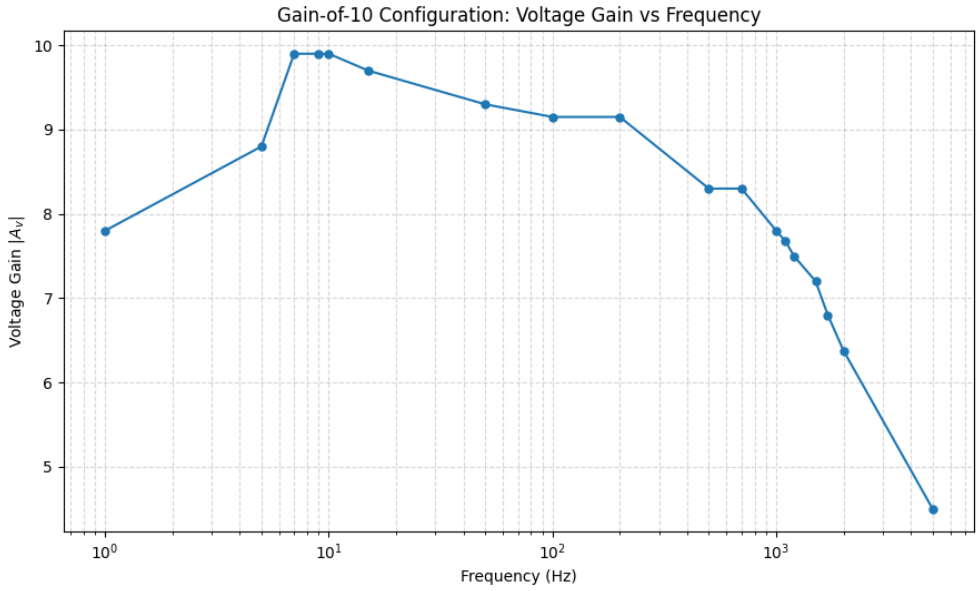


Figure 9: Frequency response curves Gain-of-10 setup.

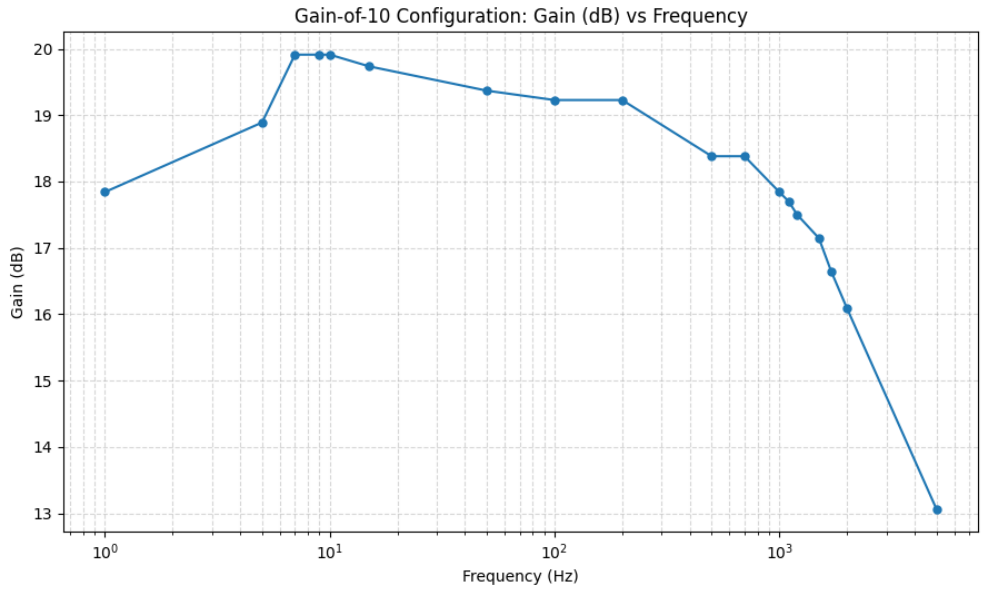


Figure 10: Frequency response curves Gain-of-10 setup(Bode Plot).

7.1 Output for 10 gain

Note : The yellow and blue curves in the oscilloscope are the V_{out1} and V_{out2} respectively (the drain voltages of the mosfets). The red/pink colour sin waves are the final $V_{out1} - V_{out2}$ to be measured(differential outputs).



Figure 11: 1.1kHz

8 Theoretical Predictions and Calculations

8.1 Open-Loop Parameter from the Previous Experiment

The measured open-loop differential gain obtained from the previous experiment is taken as

$$A_d = 56\,\mathrm{V/V}. \tag{16}$$

Its magnitude in decibels is

$$\begin{aligned} A_{d,\mathrm{dB}} &= 20\log_{10}(56) && (17)\\ &\approx 34.96\,\mathrm{dB}. && (18) \end{aligned}$$

The bias point from the previous experiment is retained, with a common-mode bias of approximately $V_{cm} = 2.2\,\mathrm{V}$.

8.2 Unity-Gain Configuration

For this configuration,

$$R_f = 18\,\mathrm{k\Omega}, \qquad R_{in} = 18\,\mathrm{k\Omega}. \tag{19}$$

8.2.1 Feedback factor

Using Equation (4),

$$\begin{aligned} \beta &= \frac{R_{in}}{R_f + R_{in}} && (20)\\ &= \frac{18\,\mathrm{k\Omega}}{18\,\mathrm{k\Omega} + 18\,\mathrm{k\Omega}} && (21)\\ &= \frac{1}{2} && (22)\\ &= 0.5000. && (23) \end{aligned}$$

8.2.2 Loop gain

The loop gain is

$$\begin{aligned} T &= \beta A_d && (24)\\ &= (0.5000)(56) && (25)\\ &= 28. && (26) \end{aligned}$$

8.2.3 Ideal closed-loop gain

From Equation (8),

$$\begin{aligned} A_{v,\mathrm{ideal}} &= -\frac{18\,\mathrm{k\Omega}}{18\,\mathrm{k\Omega}} && (27)\\ &= -1.0000. && (28) \end{aligned}$$

Thus, the ideal magnitude is

$$|A_{v,\text{ideal}}| = 1.0000, \tag{29}$$

corresponding to

$$20\log_{10}(1) = 0.00\,\text{dB}. \tag{30}$$

8.2.4 Finite-open-loop-gain prediction

Using Equation (5),

$$A_{v,\text{pred}} = -\frac{1}{1+\frac{1}{28}} \tag{31}$$

$$= -0.9655172. \tag{32}$$

Therefore,

$$|A_{v,\text{pred}}| = 0.9655172, \tag{33}$$

$$G_{\text{pred}} = 20\log_{10}(0.9655172) \tag{34}$$

$$= -0.30480\,\text{dB}. \tag{35}$$

The predicted magnitude rounded to three significant figures is

$$|A_{v,\text{pred}}| \approx 0.966\,\text{V/V}. \tag{36}$$

8.2.5 Measured gain at 1 kHz

At 1 kHz, the differential input peak-to-peak voltage was taken as

$$V_{in,pp} = 200\,\text{mV}, \tag{37}$$

and the measured output peak-to-peak voltage was

$$V_{out,pp} = 224\,\text{mV}. \tag{38}$$

Hence,

$$|A_{v,\text{meas}}| = \frac{V_{out,pp}}{V_{in,pp}} \tag{39}$$

$$= \frac{224}{200} \tag{40}$$

$$= 1.1200. \tag{41}$$

The corresponding gain in decibels is

$$G_{\text{meas}} = 20\log_{10}(1.1200) \tag{42}$$

$$= 0.98436\,\text{dB} \tag{43}$$

$$\approx 0.98\,\text{dB}. \tag{44}$$

Deviation from the ideal value:

$$\%\,\text{deviation}_{\text{ideal}} = \left|\frac{1.1200 - 1.0000}{1.0000}\right| \times 100 \tag{45}$$

$$= 12.00\%. \tag{46}$$

Deviation from the finite-A_d prediction:

$$\%\,\text{deviation}_{\text{pred}} = \left|\frac{1.1200 - 0.9655172}{0.9655172}\right| \times 100 \tag{47}$$

$$= 16.00\%. \tag{48}$$

8.2.6 Maximum measured gain in the sampled low-frequency region

The measured response reaches a maximum output of $V_{out,pp} = 228\,\text{mV}$ over the sampled 5.5–9.5 Hz region. For this measurement, the differential input peak-to-peak voltage was $V_{in,pp} = 200\,\text{mV}$.

Therefore,

$$|A_{v,\max}| = \frac{228}{200} \tag{49}$$

$$= 1.1400, \tag{50}$$

and

$$G_{\max} = 20\log_{10}(1.1400) \tag{51}$$

$$= 1.138\,\text{dB}. \tag{52}$$

Relative to the finite-A_d prediction,

$$\%\,\text{deviation}_{\text{pred}} = \left|\frac{1.1400 - 0.9655172}{0.9655172}\right| \times 100 \tag{53}$$

$$= 18.0643\% \tag{54}$$

$$\approx 18.06\%. \tag{55}$$

8.3 Gain-of-10 Configuration

For this configuration,

$$R_f = 180\,\text{k}\Omega, \qquad R_{in} = 18\,\text{k}\Omega. \tag{56}$$

8.3.1 Feedback factor

Using Equation (4),

$$\beta = \frac{R_{in}}{R_f + R_{in}} \tag{57}$$
$$= \frac{18\,\text{k}\Omega}{180\,\text{k}\Omega + 18\,\text{k}\Omega} \tag{58}$$
$$= \frac{18}{198} \tag{59}$$
$$= \frac{1}{11} \tag{60}$$
$$= 0.0909091. \tag{61}$$

8.3.2 Loop gain

The loop gain is

$$T = \beta A_d \tag{62}$$
$$= \left(\frac{1}{11}\right)(56) \tag{63}$$
$$= 5.0909. \tag{64}$$

8.3.3 Ideal closed-loop gain

From Equation (8),

$$A_{v,\text{ideal}} = -\frac{180\,\text{k}\Omega}{18\,\text{k}\Omega} \tag{65}$$
$$= -10.0000. \tag{66}$$

Thus,

$$|A_{v,\text{ideal}}| = 10.0000, \tag{67}$$

and

$$20\log_{10}(10) = 20.00\,\text{dB}. \tag{68}$$

8.3.4 Finite-open-loop-gain prediction

Using Equation (5),

$$A_{v,\text{pred}} = -\frac{10}{1 + \frac{1}{5.0909}} \tag{69}$$
$$= -8.358209. \tag{70}$$

Therefore,

$$|A_{v,\text{pred}}| = 8.358209, \tag{71}$$
$$G_{\text{pred}} = 20\log_{10}(8.358209) \tag{72}$$
$$= 18.4423\,\text{dB}. \tag{73}$$

The predicted magnitude rounded to three significant figures is

$$|A_{v,\text{pred}}| \approx 8.36\,\text{V/V}. \tag{74}$$

8.3.5 Measured gain at 1 kHz

At 1 kHz,

$$V_{in,pp} = 40\,\text{mV}, \qquad V_{out,pp} = 225\,\text{mV}. \tag{75}$$

Hence,

$$|A_{v,\text{meas}}| = \frac{V_{out,pp}}{V_{in,pp}} \tag{76}$$
$$= \frac{225}{40} \tag{77}$$
$$= 5.6250. \tag{78}$$

The corresponding gain in decibels is

$$G_{\text{meas}} = 20\log_{10}(5.6250) \tag{79}$$
$$= 15.00245\,\text{dB} \tag{80}$$
$$\approx 15.00\,\text{dB}. \tag{81}$$

Deviation from the ideal gain:

$$\%\,\text{deviation}_{\text{ideal}} = \left|\frac{5.6250 - 10.0000}{10.0000}\right| \times 100 \tag{82}$$
$$= 43.75\%. \tag{83}$$

Deviation from the finite-A_d prediction:

$$\%\,\text{deviation}_{\text{pred}} = \left|\frac{5.6250 - 8.358209}{8.358209}\right| \times 100 \tag{84}$$
$$= 32.7009\% \tag{85}$$
$$\approx 32.70\%. \tag{86}$$

The large discrepancy at 1 kHz indicates that the constant $A_d = 56$ model does not fully describe the frequency-dependent behavior of the physical amplifier. The measured frequency sweep shows that the closed-loop gain decreases significantly as frequency increases.

8.3.6 Maximum measured gain in the sampled low-frequency region

The largest measured output in the frequency sweep is $V_{out,pp} = 396\,\text{mV}$ at 7–10 Hz. With $V_{in,pp} = 40\,\text{mV}$,

$$|A_{v,\max}| = \frac{396}{40} \tag{87}$$
$$= 9.9000. \tag{88}$$

The corresponding gain in decibels is

$$G_{\max} = 20\log_{10}(9.9000) \tag{89}$$
$$= 19.91270\,\text{dB} \tag{90}$$
$$\approx 19.91\,\text{dB}. \tag{91}$$

Deviation from the ideal gain:

$$\%\,\text{deviation}_{\text{ideal}} = \left|\frac{9.9000 - 10.0000}{10.0000}\right| \times 100 \tag{92}$$
$$= 1.0000\%. \tag{93}$$

Deviation from the finite-A_d prediction:

$$\%\,\text{deviation}_{\text{pred}} = \left|\frac{9.9000 - 8.358209}{8.358209}\right| \times 100 \tag{94}$$
$$= 18.4464\% \tag{95}$$
$$\approx 18.45\%. \tag{96}$$

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9 Cutoff-Frequency Analysis

9.1 Nominal RC Corner

Both configurations have

$$f_{c,\text{nom}} = 15.9155\,\text{Hz} \approx 15.9\,\text{Hz}. \tag{97}$$

9.2 Measured Low-Frequency −3 dB Point

The low-frequency cutoff should be identified relative to the measured flat or peak gain. The available data do not contain a low-frequency point that is 3 dB below the peak for either configuration.

For the unity-gain configuration, the maximum measured value is 1.14, corresponding to

$$G_{\max} = 1.138\,\text{dB}. \tag{98}$$

Therefore, the corresponding $-3\,\mathrm{dB}$ target is

$$G_{-3\mathrm{dB}} = 1.138 - 3.000 = -1.862\,\mathrm{dB}. \tag{99}$$

The lowest measured point is $-2.45\,\mathrm{dB}$ at $1\,\mathrm{Hz}$, which is already below this target.

For the gain-of-10 configuration, the peak measured value is $19.9127\,\mathrm{dB}$, so the corresponding $-3\,\mathrm{dB}$ target is

$$19.9127 - 3.0000 = 16.9127\,\mathrm{dB}. \tag{100}$$

The lowest measured point, $17.84\,\mathrm{dB}$ at $1\,\mathrm{Hz}$, is still above this target. Hence the low-frequency $-3\,\mathrm{dB}$ point lies below the measured frequency range.

Thus, the nominal $15.9\,\mathrm{Hz}$ RC value is not directly observed as a $-3\,\mathrm{dB}$ low-frequency point in the supplied measurements due to experimental real time noise and non-idealities which are expected in a real circuit.

9.3 Measured High-Frequency -3 dB Point

The high-frequency cutoff can be estimated by interpolation between the two measured points surrounding the $-3\,\mathrm{dB}$ level. Because the horizontal axis is logarithmic, interpolation is performed in $\log_{10}(f)$.

9.3.1 Unity-gain configuration

The maximum measured gain is approximately 1.14, corresponding to $1.138\,\mathrm{dB}$. Hence, the $-3\,\mathrm{dB}$ level relative to the maximum measured gain is

$$G_{-3\mathrm{dB}} = 1.138 - 3.000 = -1.862\,\mathrm{dB}. \tag{101}$$

The measured data give

$$G(32\,\mathrm{kHz}) = -2.86\,\mathrm{dB}, \tag{102}$$
$$G(34\,\mathrm{kHz}) = -3.08\,\mathrm{dB}. \tag{103}$$

Thus the measured gain-of-10 high-frequency $-3\,\mathrm{dB}$ point is approximately $34\,\mathrm{kHz}$.

9.3.2 Gain-of-10 configuration

The measured peak is $19.9127\,\mathrm{dB}$, giving the $-3\,\mathrm{dB}$ level

$$G_{-3\mathrm{dB}} = 16.9127\,\mathrm{dB}. \tag{104}$$

The adjacent measured points are

$$G(500\,\mathrm{Hz}) = 17.38\,\mathrm{dB}, \tag{105}$$
$$G(700\,\mathrm{Hz}) = 15.56\,\mathrm{dB}. \tag{106}$$

The interpolation fraction is

$$\lambda = \frac{16.9127 - 17.38}{15.56 - 17.38} = 0.25675. \tag{107}$$

Interpolating on the logarithmic frequency axis gives

$$\log_{10} f_{c,\text{high}} = \log_{10}(500) + \lambda \left[\log_{10}(700) - \log_{10}(500)\right], \tag{108}$$

$$f_{c,\text{high}} \approx 545\,\text{Hz}. \tag{109}$$

Thus the measured gain-of-10 high-frequency $-3\,\text{dB}$ point is approximately $0.55\,\text{kHz}$. There is a dip in the amplitude for high frequency because of the parasitic capacitance in the nmos during the small signal analysis.

10 Back-Calculation of the Implied Open-Loop Gain

The laboratory manual asks for a back-calculation of the open-loop gain using the measured closed-loop gain and the model in Equation (3). Let

$$m = |A_{v,\text{meas}}|, \qquad r = \frac{R_f}{R_{in}}. \tag{110}$$

Taking magnitudes in Equation (3),

$$m = \frac{A_d r}{r + 1 + A_d}. \tag{111}$$

Rearranging,

$$m(r + 1 + A_d) = A_d r, \tag{112}$$

$$m(r + 1) + mA_d = A_d r, \tag{113}$$

$$m(r + 1) = A_d(r - m), \tag{114}$$

and therefore

$$\boxed{A_d = \frac{m(r+1)}{r-m}}. \tag{115}$$

10.1 Unity-gain configuration at 1 kHz

Here,

$$m = \frac{224}{200} = 1.1200, \qquad r = \frac{18}{18} = 1. \tag{116}$$

Hence,

$$A_d = \frac{1.1200(1+1)}{1 - 1.1200} \tag{117}$$

$$= -18.6667\,\text{V/V}. \tag{118}$$

10.2 Gain-of-10 configuration at 1 kHz

Here,

$$m = \frac{225}{40} = 5.6250, \qquad r = \frac{180}{18} = 10. \tag{119}$$

Hence,

$$A_d = \frac{5.6250(10+1)}{10-5.6250} \tag{120}$$

$$= 14.1429\,\mathrm{V/V}. \tag{121}$$

Thus, the model-implied open-loop gain at 1 kHz is approximately

$$A_d \approx 14.14\,\mathrm{V/V}. \tag{122}$$

10.3 Back-calculation at the Gain-of-10 Peak

At the maximum measured gain,

$$m = 9.9000, \qquad r = 10. \tag{123}$$

Equation (115) gives

$$A_d = \frac{9.9000(11)}{10-9.9000} \tag{124}$$

$$= 1089\,\mathrm{V/V}. \tag{125}$$

This large value is a model-implied quantity corresponding to the measured closed-loop peak. It should not be interpreted as a direct measurement of the physical open-loop gain. In particular, the measured peak occurs in the very-low-frequency region, where the amplifier can exhibit substantially higher open-loop gain than at 1 kHz.

For the unity-gain peak, the maximum measured magnitude is

$$m = 1.1400, \qquad r = 1. \tag{126}$$

Equation (115) gives

$$A_d = \frac{1.1400(1+1)}{1-1.1400} \tag{127}$$

$$= -16.2857\,\mathrm{V/V}. \tag{128}$$

Again, the negative result indicates that the measured gain exceeds the ideal unity-gain magnitude, so the simple positive-A_d closed-loop model cannot provide a physically meaningful finite open-loop gain from this measurement.

11 Comparison with Theory

11.1 Midband / Peak-Response Comparison

Using the maximum measured response in the supplied data gives the following comparison.

Table 6: Comparison of measured peak response with the ideal and finite-A_d predictions.

Configuration	**Ideal** $\lvert A_v \rvert$	**Finite-**A_d $\lvert A_v \rvert$	**Measured** $\lvert A_v \rvert$	**Deviation from ideal**	**Deviation from prediction**
Unity gain	1.0000	0.9655	1.1400	14.00%	18.06%
Gain of 10	10.0000	8.3582	9.9000	1.00%	18.45%

12 Discussion of the Measured Response

The two configurations demonstrate the principal gain-setting effect expected from resistor feedback. In the peak measured region, the unity-gain arrangement reaches a magnitude of approximately 1.14 V/V, while the gain-of-10 arrangement reaches approximately 9.9 V/V. The gain-of-10 configuration therefore follows the intended resistor ratio closely, whereas the unity-gain measurement shows a larger departure from the ideal value.

The behavior at 1 kHz is notably different. The unity-gain configuration gives a measured gain of 1.1200 V/V, while the gain-of-10 configuration has fallen to 5.625 V/V. This difference is visible directly in the frequency-response data and indicates strong frequency dependence of the physical amplifier. A constant open-loop gain of 56 V/V is therefore not sufficient to predict the full measured response over the complete frequency range.

The nominal RC calculation gives 15.9 Hz for both configurations because the RC time constant used in the setup is the same for both configurations. The measured data do not exhibit a clean, directly identifiable −3 dB low-frequency crossing at that frequency. In the gain-of-10 configuration, the measured response reaches its maximum around 7–10 Hz and then decreases as frequency is increased. The measured high-frequency roll-off can be quantified directly for the gain-of-10 configuration, giving approximately 0.55 kHz. For the unity-gain configuration, the available high-frequency points are already below the −3 dB level relative to the measured peak, so the corresponding crossing cannot be reliably interpolated from the supplied data.

Several physical effects can contribute to the departure from the simple constant-A_d model. The MOSFET differential pair has frequency-dependent small-signal gain because of device capacitances such as C_{gs} and C_{gd}, while the breadboard and measurement setup introduce additional stray capacitance. The feedback network also loads the drain nodes, and the nominal ±5% resistor tolerances can alter the two feedback paths from their ideal matched values. These effects become increasingly important when the loop gain is no longer large at higher frequencies.

13 Precautions

The following precautions were observed for the circuit configuration:

- The previous experiment's DC bias network was kept unchanged when the feedback network was added.
- The input level was kept small enough to avoid clipping at the differential output.
- Component voltage ratings of $V_{DS} \leq 20\,\mathrm{V}$ and $|V_{GS}| \leq 12\,\mathrm{V}$ were respected.
- When electrolytic coupling capacitors were used, their polarity was chosen according to the DC potential across each capacitor.
- Oscilloscope ground clips were kept on the common circuit ground.
- Power was removed before changing between the unity-gain and gain-of-10 resistor/capacitor sets.
- The function-generator DC offset was maintained at 0 V unless specifically accounted for in the bias network.

14 Conclusion

The measured results show that the resistor feedback network sets the closed-loop differential gain close to the intended values, particularly for the gain-of-10 configuration. In the maximum measured region, the unity-gain configuration reached 1.1400 V/V, while the gain-of-10 configuration reached 9.9000 V/V. These values differ from the ideal resistor-ratio gains by 14.00% and 1.00%, respectively.

Using the open-loop gain of 56 V/V obtained from the previous experiment, the finite-loop-gain model predicts closed-loop magnitudes of 0.9655 V/V and 8.3582 V/V for the unity-gain and gain-of-10 configurations, respectively. The measured peak values differ from these predictions by 18.06% and 18.45%, respectively. At 1 kHz, the measured gains are 1.1200 V/V and 5.625 V/V, demonstrating the increasing importance of the frequency-dependent open-loop gain and other non-ideal effects.

The component-based RC calculation gives the same nominal corner of approximately 15.9 Hz for both configurations because the RC time constant used in the setup is the same. The supplied measurements do not provide a directly identifiable low-frequency $-3\,\mathrm{dB}$ crossing at this nominal value. The measured gain-of-10 high-frequency $-3\,\mathrm{dB}$ crossing is approximately 0.55 kHz, while the available unity-gain high-frequency points do not bracket the corresponding $-3\,\mathrm{dB}$ level relative to the measured peak. Overall, the measurements demonstrate the intended gain-control effect of negative feedback.